Making Instacart agent-ready

Response to the six scenarios. Product lead, agent-readiness engagement. Evidence current to September 29, 2026. Bracketed numbers point to the source list at the end.

The whole argument in one paragraph

Agents will not fail at Instacart because they cannot read a catalog. They will fail at the moments where a person used to be quietly deciding something: which oat milk when the first one is gone, whether "2" means two bags or two kilos, whether this sixth order of the month is really wanted. So the plan is to find those moments, write the decision down in a form a machine can read, and let autonomy grow only as fast as the evidence about those decisions does. Instacart already owns most of the paperwork for this: replacement instructions, preference prompts, an order-issue metric that counts bad replacements. The work is to hand it to an agent safely.

Question	Answer	Deciding evidence
1. Why groceries	It is a standing order with a rulebook already on file, and a refund path when it goes wrong.	Cohorts order 2.1x a month in year one and 3.9x by year six [13]; three replacement modes persist across orders [10]
2. Board demo	Build a demo, not the throwaway feature. Run it on the real substitution contract. Cap the theatre at 5 engineer-days.	Chat-window checkout converted at about a third of click-out at Walmart [17]
3. Checkout or substitution	Substitution. It is the only step with a judgment call and a running clock.	Public MCP exposes two tools, neither touches replacements [9]
4. "quantity: 2"	The ambiguity is in the contract and in a silent server-side choice. Fix with typed measures, an expected total, and a read-back.	Instacart already deprecated bare quantity and unit for a measurements array [7]
5. Order threshold	One autonomous order per rolling 7 days, ceiling of 5 a month, plus spend and new-item rules. Start in draft mode.	Weekly ritual lands 4 or 5 times a month; mature customers average 3.9 [13]
6. "Should we be worried?"	Yes, about three named things. Sign tripwires now.	Walmart changed course on in-chat checkout in March 2026 [17]

One fact frames everything below. Instacart's agent integrations today, including the one announced with Gemini, build a real cart and then send the person to Instacart to place the order [1]. The brief asks for an agent that places the order itself, within limits. That is a step past anything shipped, so it deserves to be treated as a new capability with its own evidence, not as an extension of the chat integrations.

1. Why groceries are the right place to start

The claim I would defend to the board is narrower than "AI is good at shopping", and it can be tested.

The job repeats, and it repeats more the longer someone stays. In Instacart's own cohort data for 2017 to 2022, monthly active orderers ordered 2.1 times a month in their first year and 3.9 times a month by year six [13]. Customers outside that group averaged 11 orders a year in 2024 [14]. Instacart's assistant already takes requests like "order my usuals" [15]. The recurring basket is the natural first job.

The rulebook is already written. A customer can already tell Instacart what to do when an item is out: name up to three backups, accept a shopper's best match, or take a refund, and that choice carries to later orders [10][11]. Item preferences are saved and the customer is prompted to review them every 30 days [11]. On the shopper side, the app recommends a replacement or a refund based on those stated preferences [12]. An agent does not have to invent a relationship with a personal shopper. It has to be handed the one the customer already has.

Failure has a floor, and Instacart already counts it. A wrong item becomes a refund. But this is not a trivial category: perishables are two-thirds of orders [5], and Instacart tracks an order issue ratio that includes damaged, missing, wrong and poor-replacement items [4]. That matters because we can measure agent orders against the same yardstick as human ones from day one, which most agent domains cannot.

A screen for choosing domains: Repeat, Rulebook, Reversal, Record

Frequency alone does not make a domain agent-ready. I score candidates on four questions. Does the job repeat? Does a rulebook for the customer's preferences already exist? Can a mistake be undone cheaply? Is there a record to measure against? The ratings below are my judgment, not measured data.

Domain	Repeat	Rulebook exists	Reversal	Record
Weekly groceries	High	High: three replacement modes, saved preferences	High: line-item refund	High: 1.6B lifetime orders [3]
Household reorders	High	Medium	High	Medium
Clothing	Medium	Low: fit and taste are hard to state	Medium: returns	Medium
Flights	Low	Low	Low: change fees	Medium

The honest counterweight. Consumers are not yet asking for this. Dunnhumby found 29% of surveyed US and UK shoppers excited about agentic shopping, with nearly half calling it intrusive or unnecessary [16]. So the first wedge is opt-in, limited to people who already reorder the same basket. Wedge metrics: share of recurring baskets accepted without edits, edits per order, and agent order issue ratio against the human baseline.

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2. The six-week board demo

Build a demo. Decline the feature that will not survive.

I would tell the product team plainly: I think the feature you asked for will be discarded once real agent traffic reaches it, so let me give the board something that is still standing in month four. That is a better pitch than "we are delaying", and it keeps the relationship with the team intact.

The flashy moment in agent commerce is the agent tapping buy inside a chat window. It is also the least proven. In March 2026 Walmart said purchases completed inside ChatGPT converted at about a third of the rate of sending shoppers to its own site, and called the experience unsatisfying [17]. I am not claiming that this predicts Instacart's result. I am saying the most demoable step is not the same thing as the step that works.

The salvage test

Before any demo component is built, write one line on what remains in the codebase in month four. Every component goes in one of three bins. **Keeps**: it is a real contract or service. **Wraps**: scripted, but calling a real contract. **Throws**: theatre. Two rules follow. Throws are capped at 5 engineer-days in total, and no

production contract is changed for the demo's sake. Both numbers are my starting assumptions and the product team should argue with them.

What I would put on stage: the stockout scene

A family's recurring order is placed. Midway through shopping, the oat milk is gone. The agent reads the customer's standing instruction, takes the second named backup because the first is also out, stays inside the price limit, and writes a decision line the board can read: what it saw, what rule applied, what it chose. Everything on that screen is real except the inventory feed, which is scripted. This is the same policy contract as the four-week build in question 3, so the demo cannot be built before the contract exists. That ordering is the argument: weeks 1 to 4 build the contract, weeks 5 to 6 stage it. A slide on the demo should say which parts are scripted. Boards remember being told.

If the team still insists on the original feature: ask them to sign the salvage test with it. Push back further only if it requires changing a production contract.

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3. Two engineers, four weeks: substitution, not checkout

Checkout is a permissions problem. Substitution is a decision problem, and only one of them has a clock.

I map every moment in a weekly order where a person is pulled in, then ask which of those moments an agent cannot cover today.

Stage	What the person does today	What is hard for an agent	Clock?
Build the cart	Picks or re-buys items	Low. UPCs are prioritised when supplied [8]	No
Checkout	Confirms slot and payment	Deterministic. Partner flows hand this step to Instacart today [1]	No
During shopping	Approves or declines a replacement, or relies on a saved instruction [10][11][12]	Highest. Live stock, partly recorded preferences, a person in the aisle waiting	Yes
After delivery	Reports a problem, gets a refund	Medium. Poor replacement already counts as an order issue [4]	No

Two things make the middle row decisive. First, an agent today has no way to state or read a customer's replacement policy: the public MCP server exposes two tools, create-recipe and create-shopping-list [9]. Second, Instacart is already investing in this exact spot, with a new replacement model that weights dietary preferences such as gluten-free or allergen-free [2][6]. The agent contract should expose that model's behaviour, not compete with it. Checkout still needs hardening, but hardening can follow. An unreadable decision cannot be patched later, because by then the agent has already improvised.

The four weeks

- **Week 1.** Audit what exists, then write the standing-instruction spec from the three modes already in the product, adding delegation limits: maximum price difference, same brand only, dietary tags must match, never substitute in a named category. I do not have sight of Instacart's internal APIs, so this audit is a stated assumption and the first task.
- **Week 2.** Let an agent read and write that instruction for every item in a recurring basket. Each item resolves to one of: a named backup, best match with constraints, or refund.

- **Week 3.** A decision log per replaced item: the instruction used, the options seen, the choice, a reason code. Add a timeout rule: if nobody answers inside the shopper's wait window, follow the instruction, and if there is none, refund.
- **Week 4.** Sandbox scenarios: all three backups out; a backup priced 40% above the original; the agent silent; two agents writing conflicting instructions; instruction edited mid-shop; an allergen mismatch.

Done means an agent can set instructions for every item in a recurring basket, every replacement outcome maps to a logged reason, and the scenarios above pass. After launch I would compare poor-replacement reports on agent-set instructions against the human baseline [4].

What would change my mind: if there is no scoped way to give an agent payment authority at all, checkout comes first, because nothing else can launch without it.

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4. "quantity: 2" read two ways

Where the problem lives, and how I would remove it.

There are four places this can come from. In order of how likely I think each is:

Where	What goes wrong	Evidence
1. The contract	A bare number with an optional unit says nothing about whether it means items, packs, or weight.	Instacart deprecated the quantity and unit fields on LineItem in favour of a line_item_measurements array [7]
2. The resolver	Given several measurements, Instacart chooses the best one to compute quantity. The agent never sees which, so two platforms can send the same payload and expect different carts.	Same changelog entry [7]
3. The catalog	"Wheat flour 1 kg" can be a 1 kg pack, a loose-weight item, or a multipack. Same words, different objects. Units outside the supported list fail matching.	Docs on units and matching [8]
4. Each agent's parsing	How a model turns "two bags of atta" into JSON. Least controllable, so we should not rely on it.	Inference, not sourced

The fact that Instacart moved off a bare quantity on its own endpoint is the tell: the platform already learned that a single number is not enough for people. Agents make the same flaw louder. In my earlier draft I argued against deprecating the integer where the catalog supplies context. Given this changelog I have reversed that: the agent-facing path should not accept a bare quantity at all.

The fix in three layers

Typed measure. The agent-facing schema requires a kind (each, weight, volume, pack), a value, and a unit from the supported list [8]. Identity is by UPC where possible, which Instacart already searches on exclusively when present [7].

Expected total. The request also says what the customer expects to receive, for example "2 kg of flour, any pack mix" or "2 packs". The platform checks its resolution against it and rejects or flags a mismatch, the way a checksum catches a mistyped number.

Read-back. The response returns the platform's interpretation in structured, plain terms, such as "resolved: 2 x 1 kg pack, 2.0 kg total". Aviation calls this a read-back. Any agent, and the audit log, can see what the

platform understood, so disagreement surfaces at order time instead of at the door.

Testing. One shared corpus of about 100 real listings, deliberately including multipacks, loose produce and dual-unit items, run by every integrated platform. A failing case blocks release. Track the rate at which expected total and resolved total differ, and add a support tag for wrong-quantity complaints. The corpus size is a proposal.

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5. How many autonomous orders per user per month

My answer is a rule set with one starting number, and a way to replace that number with evidence.

Launch position: one autonomous order per rolling 7 days, a ceiling of 5 per calendar month, and a spend limit the customer sets. I would not describe 5 as an estimate of risk, because there is no agent-order history to estimate from. It is a ceiling chosen so a normal shopper never meets it, which makes hitting it a signal.

Where the number comes from

A weekly shopper orders on 4 or 5 dates in any month. Instacart's most mature cohorts averaged 3.9 orders a month [13], and the less frequent group averaged 11 a year [14]. Five sits above both averages and matches a weekly ritual with no room for extras. The cohort figures are from 2017 to 2022 cohorts and averages hide heavy users, so anyone who genuinely needs more can raise their own ceiling.

The rolling-week rule does more work than the monthly one. A monthly cap lets a runaway loop place five orders in an afternoon. A weekly window stops it at the second.

The other limits

- **Spend.** Auto-approve up to a ceiling the customer sets at enrolment. I would suggest a default near the platform average order value of $115 [2] and ask for confirmation when an order exceeds 1.25 times the customer's median of their last five orders. Both are pilot values.
- **Nothing new unseen.** An item the customer has never bought needs a tap. Anything in the cart should be something they would recognise on the receipt.
- **Substitutions** only within the standing instruction from question 3.

Getting data when there is none

The tension in the case is real: agent orders will raise volume and generate "I didn't mean to order this" complaints, with no history to forecast either. So each customer climbs three rungs.

Rung	How it works	What it teaches us
1. Draft	The agent builds the basket. The customer presses send.	How often drafts get edited, and what changes
2. Send with hold	The agent places the order, held before the shopper starts, cancellable in one tap. The hold length is an operations decision.	Cancellation rate
3. Autonomous	Inside the limits above.	Issue ratio in the wild

A customer or cohort moves up a rung only when agent orders match the human baseline on Instacart's existing order issue ratio, defined as orders with at least one issue over total orders and including poor replacement [4], over an agreed number of orders. I would also add "didn't mean to order this" as a reason code in the complaint flow, so the exact complaint in the case becomes a counted metric. The north-star

measure is agent orders delivered with no edit, no issue and no such tag. Enrolment is opt-in only, for the reasons in question 1 [16].

6. “Should we be worried? Will this happen to us?”

Say what we know, what we do not, and what would make us stop. Then put it in writing.

What I would say to the client contact:

> *“Yes, be worried, but about specific things and not about the whole category. A competitor pausing after four months tells us this is hard to execute. It does not tell us why theirs stopped, and I am not going to guess about their project in front of you.*
>
> *The closest public case I can point to is Walmart’s in-chat checkout, which converted at about a third of the rate of its own site before Walmart changed course [17]. That was less about a model getting a shelf wrong and more about where the purchase happens and who holds the customer’s trust.*
>
> *Three things could go the same way for us. We could ask customers to accept unreviewed orders before they are ready. The agent could misread what someone meant, on quantity or on a substitution. Or nobody could explain afterwards why it did what it did. Here is what we are doing about each, here is what we cannot yet know, and here are the three conditions under which we stop. I would like us both to sign them today.”*

Our own uncertainty, stated plainly

We know	We do not know	When we will know
Order quality has improved for 16 straight quarters [3]; assistant orders run larger than the $115 average [2]; integrations already exist with Claude, ChatGPT and Gemini [1]	How often an agent-placed order is one the customer did not mean. There is no history. We also cannot see the internal state of Instacart’s APIs, and appetite for unreviewed ordering is unproven [16]	After rung 1 (draft mode): four weeks of edit data give the first real estimate

Three tripwires, with numbers set in week 2 against the human baseline

- Agent order issue ratio worse than the human baseline by an agreed margin over an agreed volume: freeze promotion between rungs.
- Unauthorized-order complaints above an agreed rate: return that cohort to draft mode.
- Any quantity read-back mismatch that reaches a shopper: release blocker.

On the client blaming the consulting team elsewhere. I would not distance us from the other firm or criticise it. The protection for both sides is paper: a one-page assumptions log recording what we believed, on what evidence, and who decided, plus the signed tripwires. If a decision goes wrong, the log shows whether it was made with the evidence available at the time.

Appendix A. What changed from the earlier draft

- Dropped the 70 to 80 percent same-SKU claim, the 65.2 percent repeat-intent figure, the DABstep accuracy figure, the stockout and inventory-distortion figures, and the Visa dispute thresholds. None was needed for a decision, and most were not Instacart data. Replaced them with Instacart's own cohort, order-quality and product-documentation data.
- Reversed the quantity position: Instacart's changelog shows it already deprecated bare quantity on the shopping-list endpoint, so the agent path should not accept one.
- Rebuilt the order threshold around a rolling week and a draft-first ladder, and stated where 5 comes from instead of presenting it as an optimum.
- Replaced the competitor script, which described a failure mode the case does not establish, with a public precedent and a set of tripwires.

Appendix B. Evidence limits

Read these numbers with the following in mind. The 2.1x to 3.9x cohort figures come from a Q3 2023 earnings call as summarised on a third-party transcript page, cover 2017 to 2022 cohorts, and are marketplace-wide, not agent-specific [13]. The 11-orders-a-year figure is from a 2024 report of an executive's remarks [14]. Walmart's one-third figure was first reported by Wired and relayed here via Search Engine Land [17]. The Dunnhumby figures are relayed through Grocery Dive [16]. Product behaviour described from Instacart documentation may have changed since the pages were indexed. The four-week plan assumes internal APIs I have not seen. Every numeric threshold in this document (5 per month, 1.25 times median basket, $115 default, 5 engineer-days, 100-listing corpus) is a proposal to tune, not a finding.

Appendix C. Sources

[1] Instacart, "Instacart Brings Agentic Grocery Shopping to Gemini", May 19, 2026
https://company.instacart.com/updates/instacart-brings-agentic-grocery-shopping-to-gemini

[2] Instacart, Second Quarter 2026 Financial Results, Aug 6, 2026
https://investors.instacart.com/news-releases/news-release-details/instacart-announces-second-quarter-2026-financial-results

[3] Instacart, Q2 2026 Prepared Remarks, Aug 6, 2026
https://investors.instacart.com/static-files/e052f585-2444-4c73-8687-a73efadc9820

[4] Instacart, Q2 2026 Supplemental Information (order issue ratio, perfect-order fill rate definitions)
https://investors.instacart.com/static-files/507a262a-6a9c-45a5-88db-7dcafd0b5f15

[5] Investing.com, Instacart Q2 2026 slides (perishables two-thirds of orders)
https://www.investing.com/news/company-news/instacart-q2-2026-slides-revenue-up-14-cash-flow-surges-156-93CH-4844880

[6] Investing.com, Instacart Q2 2026 earnings call transcript (replacement model)
https://m.uk.investing.com/news/transcripts/earnings-call-transcript-instacart-q2-2026-results-send-shares-up-93-93CH-4818029?ampMode=1

[7] Instacart Docs, Developer Platform changelog (quantity and unit deprecated; line_item_measurements; UPC search)
https://docs.instacart.com/developer_platform_api/api/changelog/

[8] Instacart Docs, Create shopping list page (units, UPC priority)
https://docs.instacart.com/developer_platform_api/api/products/create_shopping_list_page

[9] Instacart Docs, Connect your AI agent to Instacart with MCP (create-recipe, create-shopping-list tools)
https://docs.instacart.com/developer_platform_api/guide/tutorials/mcp

[10] Instacart Docs, Replacements (three modes, saved backups)
https://docs.instacart.com/storefront/learn_about_your_storefront/cart_and_checkout/replacements/

[11] Instacart Help Center, adding item and replacement instructions (carry over; 30-day preference review)
https://www.instacart.com/help/section/360007902831/360039162252

[12] Instacart shopper community, "Providing a more straightforward replacements experience", Oct 2025
https://company.instacart.com/shopper-community/providing-a-more-straightforward-replacements-experience

[13] Maplebear Q3 2023 earnings call, via MarketBeat (cohort order frequency 2.1x to 3.9x)
https://www.marketbeat.com/earnings/reports/2023-11-8-instacart-maplebear-inc-stock

[14] PYMNTS, Instacart converting casual users into regulars, Sept 2024 (11 orders a year)
https://www.pymnts.com/news/delivery/2024/instacart-aims-to-cement-online-grocery-dominance-by-converting-casual-users-into-regulars/

[15] PYMNTS, Instacart launches Clementine, Sept 9, 2026
https://www.pymnts.com/news/artificial-intelligence/2026/instacart-launches-ai-agent-clementine-to-turn-chats-into-grocery-carts/

[16] Grocery Dive, “The Friday Checkout: Is agentic AI the next frontier for grocers?”, Mar 2026 (Dunnhumby)
https://www.grocerydive.com/news/the-friday-checkout-agentic-ai-next-frontier-grocery-technology-instacart-walmart/818407/

[17] Search Engine Land, Walmart: ChatGPT checkout converted 3x worse than website, Mar 2026
https://searchengineland.com/walmart-chatgpt-checkout-converted-worse-472071